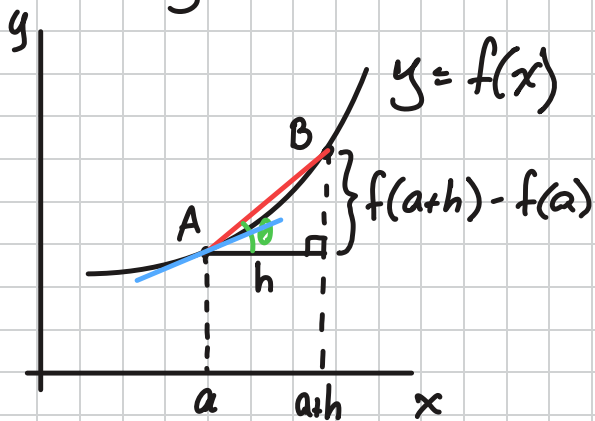


Week 4

derivatives

tangent lines



$$m = \tan \theta = \frac{f(a+h) - f(a)}{h} \quad \text{Slope of Secant AB}$$

to get tangent line at $x=a$
we take the limit as $h \rightarrow 0$

$$\text{Slope of tangent line: } m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Example

$f(x) = \sqrt{x}$ Find eq of tangent line to $f(x)$ @ $a=4$

$$m = \lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h} \times \frac{\sqrt{4+h} + 2}{\sqrt{4+h} + 2}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{4+h} - 4}{\cancel{h}(\sqrt{4+h} + 2)}$$

$$= \frac{1}{\sqrt{4+0} + 2} = \left(\frac{1}{4} \right)$$

now find point

$$y = \sqrt{x}$$

$$y = \sqrt{4}$$

$$= 2 \quad P(4, 2)$$

$\therefore$ at $P(4, 2)$ Slope is $\frac{1}{4}$

$$y - y_0 = m(x - x_0)$$

$$y - 2 = \frac{1}{4}(x - 4)$$

$$\Rightarrow y = \frac{x}{4} + 1$$

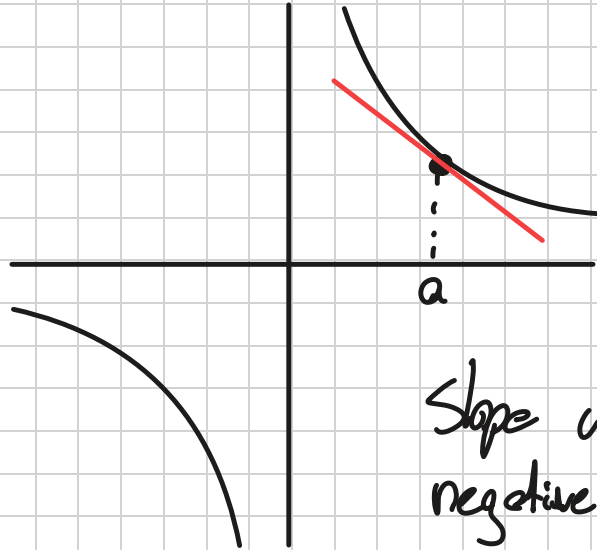
② find equation of normal line

$$m_n = -4$$

$$y - 2 = -4(x - 4)$$

$$y = -4x + 18$$

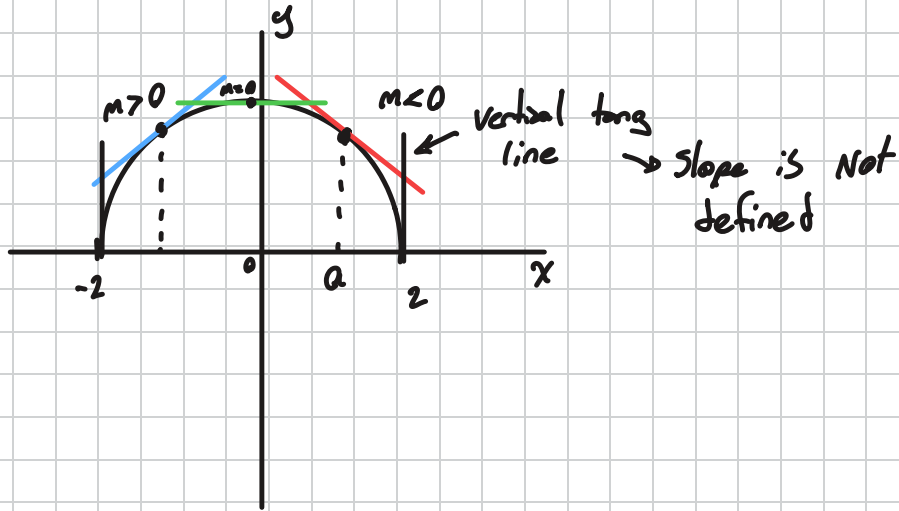
$$y = -\frac{1}{2^2}x + \frac{2}{a}$$



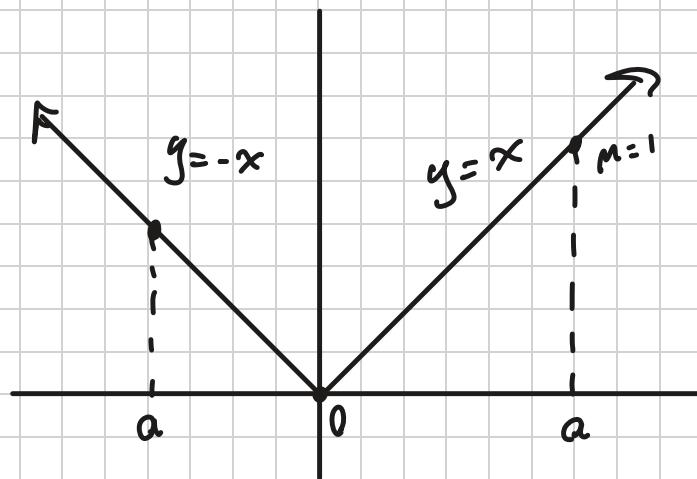
Slope will be
negative everywhere

Ex $f(x) = \sqrt{4-x^2}$ $4-x^2 \geq 0$ or $x^2 \leq 4$
 $\Rightarrow x \in [-2, 2]$

$y = \sqrt{4-x^2} \Rightarrow y^2 = 4-x^2$
 $\Rightarrow x^2 + y^2 = 4$ equation of Circle



Ex $f(x) = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$



No tang line
at $x=0$

Differentiable

$f(x)$ domain D

Def f is differentiable at $x \in D$
if $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exists and is limit

then $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

the derivative of $f(x)$ at x

if f is differentiable at all $x \in D$

$\Rightarrow f'(x)$ is also a function on D

$$E_x \textcircled{1} f(x) = x^2$$

$$D(f) = \mathbb{R}$$

$$f'(x) = 2x$$

$$D(f') = \mathbb{R}$$

$$\textcircled{2} f(x) = \frac{1}{x}$$

$$D(f) = \mathbb{R} - \{0\}$$

$$f'(x) = -\frac{1}{x^2}$$

$$D(f') = \mathbb{R} - \{0\}$$